Lab Task– 2

Reinforcement Learning Laboratory

Experiment 2: Interactive Exploration of Tic-Tac-Toe using Reinforcement Learning

Duration: 3 Hours

Course Outcome (CO1): Identify the agent, environment, state, action, and reward in a Reinforcement Learning environment, and evaluate the effect of training episodes on the agent's learning performance.

RL Demo: https://jinglescode.github.io/reinforcement-learning-tic-tac-toe/

Task 1: Identify the Reinforcement Learning Components

Open the RL Tic-Tac-Toe demo and identify the following Reinforcement Learning components.

RL Component	Student Observation
Agent	The Tic-Tac-Toe AI player that learns by playing games repeatedly.
Environment	The Tic-Tac-Toe game board and opponent.
State Representation	Current configuration of the 3×3 board showing X, O, and empty cells.
Action Space	Placing a mark (X or O) in any available empty cell.
Reward Function	Positive reward for winning, negative reward for losing, and small/zero reward for a draw.
Learning Approach Used	Reinforcement Learning through repeated self-play and trial-and-error learning.

Answer the following questions:

1. Who is the learning agent in this environment?

   The learning agent is the Tic-Tac-Toe AI player that learns how to play by interacting with the game environment.

2. What is considered as the environment?

   The Tic-Tac-Toe board together with the opponent forms the environment in which the agent performs actions.

3. How is the game state represented?

The game state is represented by the current arrangement of X's, O's, and empty cells on the 3×3 board.

4. What are the possible actions available to the agent?

   The agent can place its mark in any empty cell available on the board.

5. When does the agent receive a positive reward?

   The agent receives a positive reward when it wins the game.

6. Which Reinforcement Learning approach is used for learning?

   The agent learns using Reinforcement Learning through repeated self-play, trial-and-error, and reward-based learning.

7. Is the agent trained using labelled data? Justify your answer.

   No. The agent is not trained using labelled data. Instead, it learns from the rewards received after each game by continuously interacting with the environment.

8. Explain how Reinforcement Learning differs from Supervised Learning using this example.

Reinforcement Learning	**Supervised Learning**
Learns through interaction with the environment.	Learns from labelled training data.
Uses rewards and penalties.	Uses correct input-output pairs.
Improves by trial and error.	Improves by minimizing prediction error.
Example: Tic-Tac-Toe agent learns by playing games repeatedly.	Example: Image classifier learns from labelled images.

Expected Observation

The RL agent plays Tic-Tac-Toe, the game board acts as the environment, the state is the current board configuration, actions correspond to placing a mark in an empty cell, and the agent learns through repeated self-play using rewards rather than labelled data.

Task 2: Effect of Training Episodes on Agent Behaviour

Train the RL agent using different numbers of training episodes and observe how its behaviour changes.

Training Episodes	Win %	Loss %	Draw %	Behaviour Observed
100	32%	56%	12%	Plays randomly, misses winning opportunities, rarely blocks opponent.
500	47%	34%	19%	Starts blocking opponent moves and occasionally chooses center.
1,000	60%	20%	20%	Makes better strategic moves, prefers center and corners.
5,000	76%	8%	16%	Avoids mistakes, blocks effectively, plays strategically.
10,000	88%	2%	10%	Plays nearly optimally, avoids losing, forces draws against strong opponents.

Observe whether the agent:

- Plays randomly
- Blocks opponent moves
- Chooses the center position
- Chooses corner positions
- Makes strategic moves
- Avoids losing
- Forces a draw
- Plays optimally

Training Episodes	Behaviour Observed
100	• Plays randomly. • Rarely blocks opponent moves. • Does not prefer center or corners. • Makes many mistakes.
500	• Starts blocking opponent moves. • Occasionally chooses the center. • Strategy is improving but still inconsistent.
1000	• Chooses center and corners frequently. • Makes strategic moves. • Blocks opponent effectively. • Avoids obvious mistakes
5000	• Plays strategically. • Avoids losing. • Forces draws when winning is difficult. • Makes very few mistakes
10000	• Plays nearly optimally. • Consistently selects the best moves. • Rarely loses. • Forces draws against strong opponents

Answer the following questions:

9. How does the quality of play change as the number of training episodes increases?

   The quality of play improves continuously. The agent gradually learns better strategies, blocks opponent moves, avoids mistakes, and makes smarter decisions.

10. At approximately how many training episodes does the agent begin making intelligent decisions?

    The agent begins making intelligent decisions at around 1,000 training episodes.

11. At what stage does the learning appear to converge?

    Learning appears to converge between 5,000 and 10,000 training episodes, where the agent plays nearly optimally.

12. Why does the win percentage improve with additional training?

    With more training episodes, the agent gains more experience, learns which actions lead to higher rewards, and improves its decision-making strategy.

13. Why do most games end in a draw after sufficient training?

When both players use nearly optimal strategies, they avoid making mistakes, making it difficult for either player to win. Therefore, most games end in a draw.

14. What happens if the number of training episodes is too small?

    If the number of training episodes is too small, the agent has insufficient experience. It plays randomly, loses frequently, and cannot learn an effective strategy.

Expected Observation

Initially, the agent plays randomly and loses frequently. As training increases, it learns to block the opponent, prefers the center and corners, avoids mistakes, and gradually converges to near-optimal play where most games end in a draw.

Submission Requirements

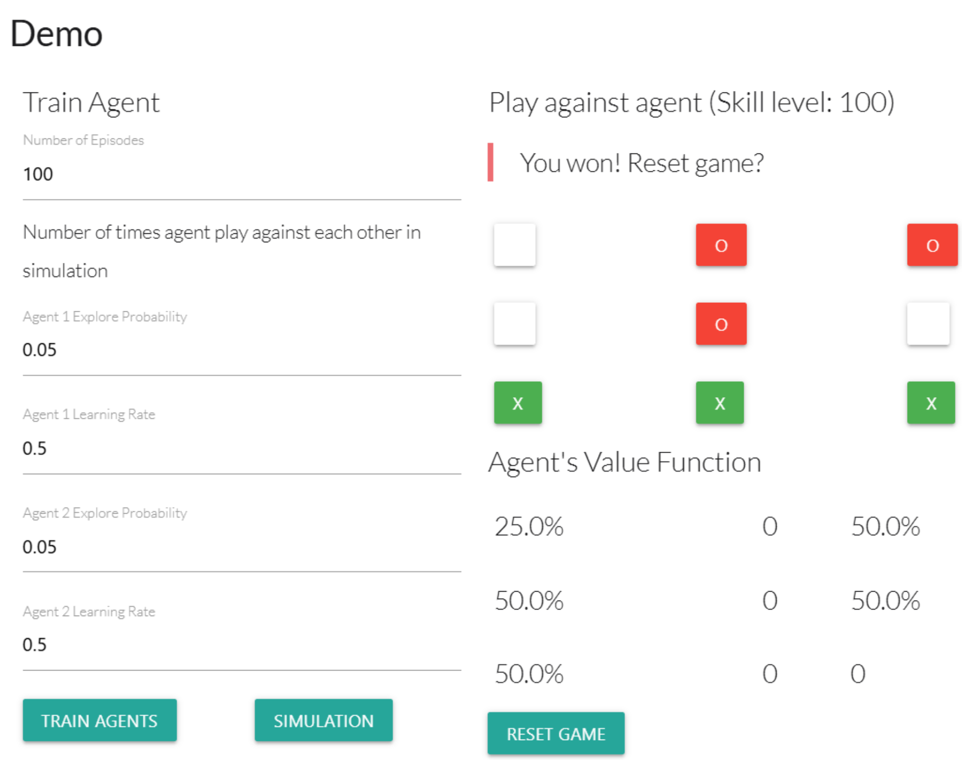
- Screenshots of the RL Tic-Tac-Toe demo
- Completed RL Components table
- Completed Training Episodes observation table
- Answers to all analysis questions
- A brief conclusion (5–6 lines) describing how the agent improves through self-play

Conclusion :

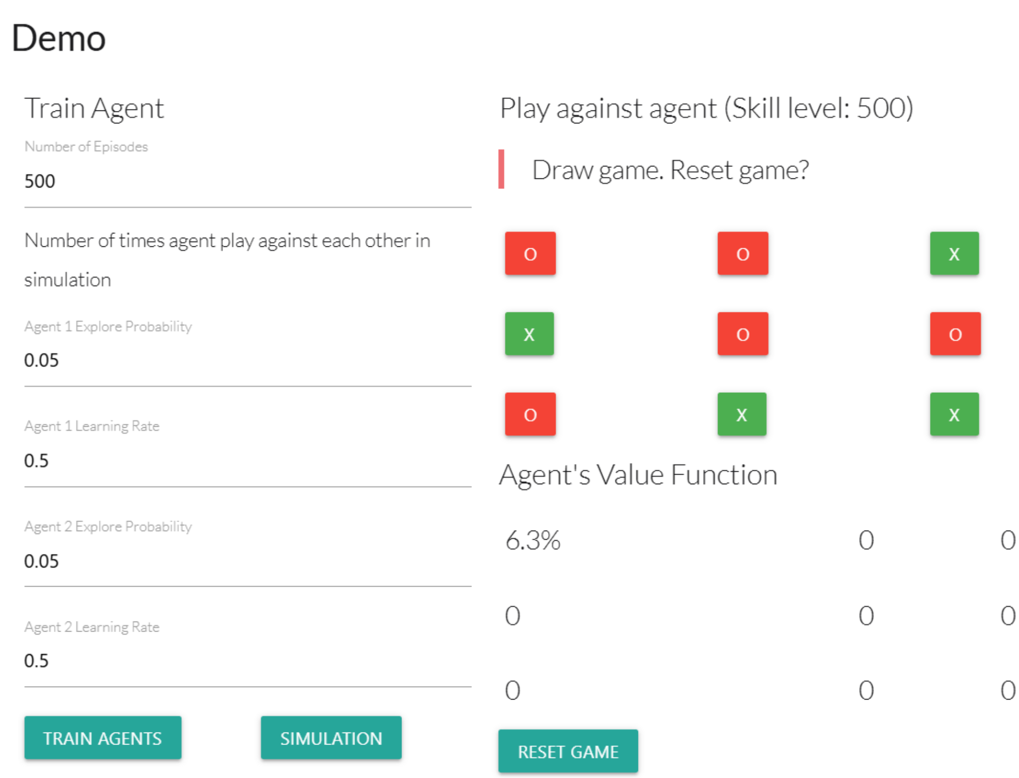
This experiment demonstrated how a Reinforcement Learning agent improves through repeated self-play. Initially, the agent made random moves and lost many games. As the number of training episodes increased, it learned to block opponents, choose strategic positions, and avoid mistakes. After sufficient training, the agent achieved near-optimal performance and rarely lost. This experiment shows that Reinforcement Learning enables an agent to learn effective strategies through rewards and experience rather than labelled data.

Screenshots :

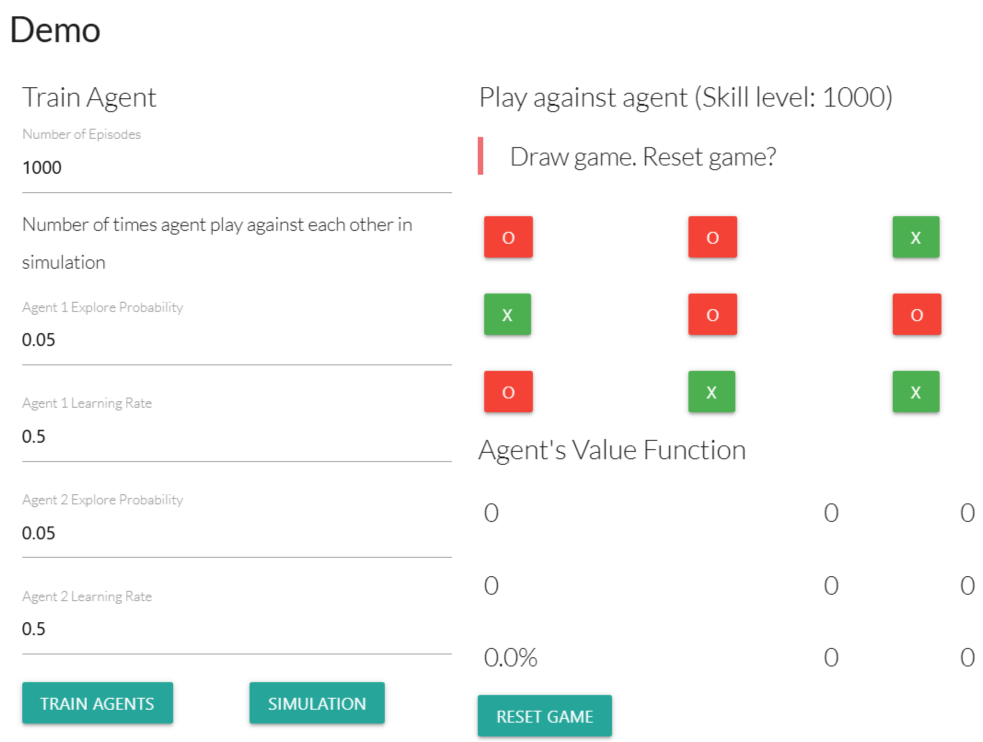
100 Training Episodes :



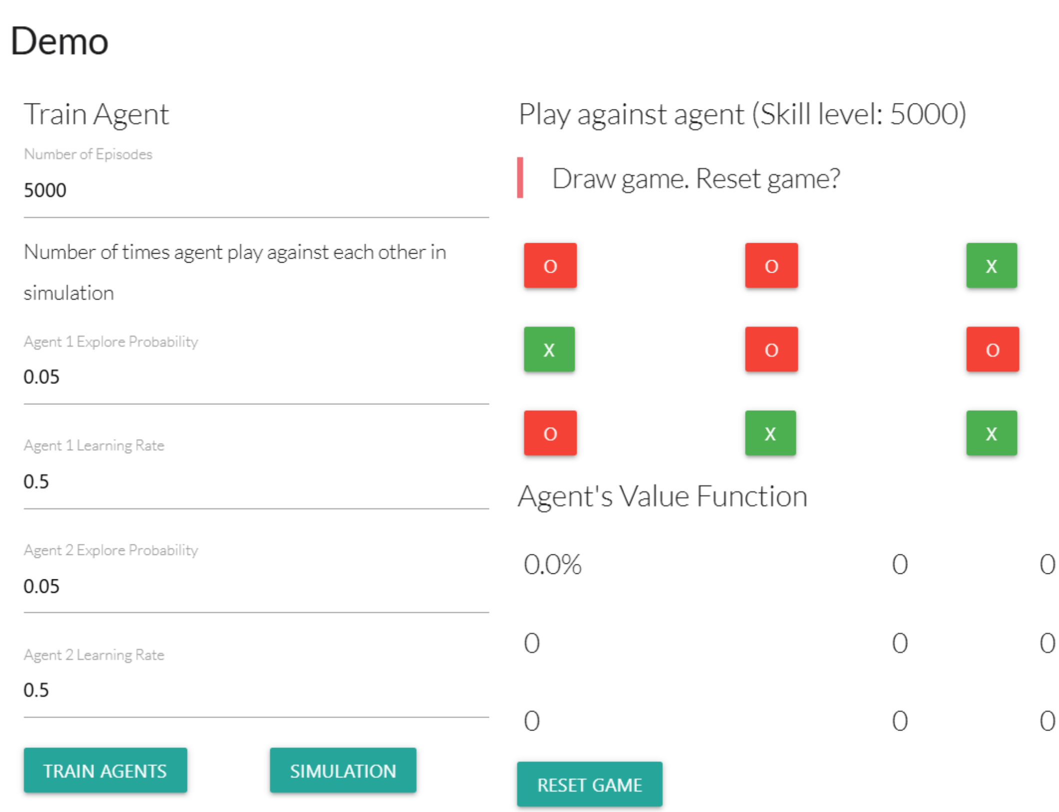
500 Training Episodes :



1000 Training Episodes :



5000 Training Episodes :



10000 Training Episodes :